Figure 13.2 **(**a**)** Open strings with endpoints attached to two-dimensional branes, or two-branes. **(**b**)** Strings stretching from one two-brane to another. **(**c**)** Strings stretching from a two-brane to a one-brane.

That was a paramount result. It showed that decades of research that produced sharp mathematical methods to study one-dimensional objects —strings—could be used to study higher-dimensional objects, *p*-branes. Wonderfully, then, Polchinski revealed that the analysis of higher-dimensional objects was reduced, to a large degree, to the thoroughly familiar, if still hypothetical, analysis of strings. It's in this sense that strings are special among equals. If you understand the behavior of strings, you're a long way toward understanding the behavior of *p*-branes.

With these insights, let's now return to the braneworld scenario—the possibility that we're all living out our lives within a three-brane.

Our Universe as a Brane

If we are living within a three-brane—if our four-dimensional spacetime is nothing but the history swept out by a three-brane through time—then the venerable question of whether spacetime is a something would be cast in a brilliant new light. Familiar four-dimensional spacetime would arise from a real physical entity in string/M-theory, a three-brane, not from some vague or abstract idea. In this approach, the reality of our four-dimensional spacetime would be on a par with the reality of an electron or a quark. (Of course, you could still ask whether the larger spacetime within which strings and branes exist—the eleven dimensions of string/M-theory—is itself an entity; the reality of the spacetime arena we directly experience, though, would be rendered obvious.) But if the universe we're aware of really is a three-brane, wouldn't even a casual glance reveal that we are immersed within something—within the three-brane interior?

Well, we've already learned of things within which modern physics suggests we may be immersed—a Higgs ocean; space filled with dark energy; myriad quantum field fluctuations—none of which make themselves directly apparent to unaided human perceptions. So it